Sonika Joshi

Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2025

Bachelor in Computer Applications
Course Title: Computer Graphics and Animation
Code No: CACS 305
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

2. Define raster and random scan display systems. Draw and explain how each system renders a diagonal line. [2+3]
3. Discuss the importance and workflow of area filling algorithms with diagrams. Explain how the flood fill algorithm works with 4-connected and 8-connected boundaries. [1+4]
4. Derive the homogeneous matrix for reflection about the line y = x
5. Describe composite transformations. Prove that the order of transformations matters by applying rotation followed by translation vs. translation followed by rotation to a point.[1+4]
6. Describe the purpose of window-to-viewport transformation. Derive the transformation equation and explain how aspect ratio is preserved. [1+4]
7. Explain the Back-Face Removal algorithm. What are its advantages and limitations? [3+2]
8. Write short notes on: (any two)
   a) Keyframe Animation b) Application of Computer Graphics c) DDA algorithm

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. Derive the general formula for the decision parameter in Bresenham's algorithm. How does integer arithmetic help avoid floating-point operations? [7+3]
10. Create a scenario in which the Z-buffer and Painter's algorithm produce different results. Explain why the outputs differ and illustrate how the scene would be rendered using both methods. You can include diagrams or pseudocode to support your explanation. [Hint: Start with three overlapping polygons (e.g., triangles or rectangles) in 3D space such that: One is in front of another in some regions, And behind it in other regions (i.e., their Z-depths intersect or overlap).] [5+3+2]
11. What is Virtual Reality (VR)? Explain the components of a typical VR system and list its uses in modern animation and simulation. [2+8]

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T.U. Examination Question

TRIBHUVAN UNIVERSITY

Faculty of Humanities & Social Science
Office of the Dean
2020

Bachelor Level/Third Year/Fifth Semester/Science
Computer Graphics and Animation (CACS 305)
(Computer Graphic)
(New Course)

Full Marks: 60
Pass marks: 24
Time: 2 hrs.

Candidates are required to give their answer in their own words as for as practicable.

Group 'B'

Attempt any SIX questions: **(6 × 5 = 30)**

2. What is computer graphics? Explain different application areas of computer graphics.
3. How can you draw a circle using mid-point algorithm? Explain with the algorithm.
4. Explain 3D basic geometric transformation with an example.
5. What is polygon clipping? Explain Sutherland Hodgman algorithm for polygon clipping.
6. Give a triangle with vertices A(2, 3), B(5, 5) and C(4, 3) by rotation 90 degree about the origin and then translating two unit in each direction. Use homogeneous transformation matrix to find the new vertices of the triangle.
7. Explain the scan line algorithm for visible surface detection.
8. Explain the architecture of VR System with necessary components.

Group 'C'

Attempt any TWO questions: **(2 × 10 = 20)**

9. Explain the working details of DDA line drawing algorithm? Digitize the line with end points (2, 2) and (10, 5) using Bresenhman's line drawing algorithm. 5+5=10
10. Write the difference between object space method and image space method. Explain Z buffer algorithm for visible surface detection. 5+5=10
11. Derive the formula for windows to viewpoint transformation. Given a window bordered by (0, 0) at the lower left and (4, 4) at the upper right. Similarly, a viewport bordered by (0, 0) at the lower left and (2, 2) at the upper right. If a window at position (2, 4) is mapped into the viewpoint. What will be the position of viewport to maintain same relative placement as in window ? 5+5+10

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Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2025

Bachelor in Computer Applications
Course Title: MIS and e-Business
Code No: CACS 301
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

[6×5 = 30]

Attempt any SIX questions.

2. What is the fundamental difference between E-Business and E-Commerce? Provide a brief example illustrating each.
3. What is SCM? Explain in detail with proper example?
4. Considering Location-Based Services (LBS) applications, which major Barrier of EC poses the most significant challenge to widespread user adoption and trust? Justify your answer briefly.
5. Explain the architecture of WAP? Explain how WAP works?
6. Write short note on
   a) VPN b) IPS
7. What is C panel? Explain the steps to upload EC site to the web Server.
8. What is EC defense strategy? Explain PKI in detail?

Group C

[2×10 = 20]

Attempt any TWO questions.

9. What Is Business Plan, discuss its components? Explain website usability and website promotion in detail. [5+5]
10. Explain different website structure in detail. Explain the security issues related to wireless communications? [5+5]
11. What are the components of Expert System? Highlight the role of Decision Support System, executive information system and Transaction processing system in a successful company? [1+3+3+3]

Tribhuvan University

Faculty of Humanities and Social Science
O FFICE OF THE DEAN
2020

Bachelor in Computer Applications **Full Marks: 60**

Course Title: MIS and e-Business **Pass Marks: 24**
Code No.: CACS 301 **Time: 3 hours**
Semester: V

Candidates are required to answer the questions in their own words as far as possible.

Group 'B'

Attempt any six questions. **[6 × 5 = 30]**

2. How properties like ubiquity, richness and information density make e-commerce better than traditional commerce? [5]

3. Differentiate B2C, B2B and C2C e-commerce with example. [5]

4. What is the role of information superhighway in e-commerce? List the components of information superhighway. [3 + 2]

5. What is international information system? Describe the concepts of outsourcing and offshoring in the system. [1 + 4]

6. Define availability, authentication and authorization. How they can be enforced in e-commerce security? [3 + 2]

7. What is e-checking? Describe its working mechanism. [2 + 3]

8. Describe the website design criteria that one should follow while designing an e-commerce website. [5]

Group 'C'

Attempt any TWO questions. **[2 × 10 = 20]**

9. What do you mean by Wireless Application Protocol (WAP)? How it works? Discuss the layered architecture of WAP. [2 + 2 + 6]

10. How important web contents are for e-commerce system? Discuss their types. Justify, with examples, how web contents can be used to promote cross-selling, up-selling and promotions in e-commerce. [2 + 2 + 6]

11. Discuss in detail security defends strategies that you can implement while securing e-commerce systems from security attacks and threats. [10]

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T.U. Examination Question

TRIBHUVAN UNIVERSITY

Faculty of Humanities & Social Science

Office of the Dean

2020

Bachelor Level/Third Year/Fifth Semester/Science — Full Marks: 60

Dot Net Technology (CACS 302) — Pass marks: 24

Time: 3 hrs.

Candidates are required to give their answer in their own words as for as practicable.

Group 'B'

Attempt any SIX questions: **(6 × 5 = 30)**

2. Differentiate Object Oriented Programming and Object Based Programming. Explain some of the major features of C# language. 2+3
3. Explain overview of Microsoft.NET framework and its components in detail. 5
4. What do you mean by property in C# language ? How it is different from method ? Compare automatic property with other types of property with suitable example. 1+1+3
5. Define constructor. Explain different types of constructors used in C# with example. 1+4
6. Define inheritance. Write C# program to demonstrate multilevel and multiple inheritance. 1+2+2
7. What is generics ? List different types of generic classes. Explain delegate with example. 1+1+3
8. What do you mean by lambda expression? Explain different types of lambda expression used in C# with example. 1+4

Group 'C'

Attempt any TWO questions: **(2 ×10 = 20)**

9. a) Write a program to create user registration from in one ASP.NET web page and siplay filled data in another page. 7
   b) Write a program for handling exception in ASP.NET. 3
10. a) How virtual method is used to achieve dynamic binding in C#? Explain with the help suitable program. 1+4
    b) Define operator overloading. Write a C# program to overload binary operator. 1+4
11. a) What is LINQ? Write a program to select employees whose salary is grater than 20000 and whose address is Kathmandu using LINQ. 1+4
    b) Write a C# program to show insert and select operation in database. 5

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T.U. Examination Question

TRIBHUVAN UNIVERSITY

Faculty of Humanities & Social Science

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2020

Bachelor Level/Third Year/Fifth Semester/Science — Full Marks: 60

Computer Networking (CACS 303) — Pass marks: 24

Time: 3 hrs.

Candidates are required to give their answer in their own words as for as practicable.

Group 'B'

Attempt any SIX questions: **(6 × 5 = 30)**

2. Define protocols. Explain WWW and HTTP protocol. 1+2+2
3. Define transmission impairment. Explain the causes of impairments. 1+4
4. Define HDLC. Explain the HDLC frame formats. 1+4
5. Define IP Address. Specify IPv4 address classes with their address ranges. 1+4
6. Define Subnetting. Suppose you are given network address: 192.168.10.0 and subnet mask: 255.255.255.240 then calculate total number of subnets and total number of hosts per subnet. 1+2+2
7. Draw a User Datagram format. Explain UPD operation. 2+3
8. Write short notes on (Any Two): 2.5+2.5
   (a) DNS
   (b) Public Key Cryptography
   (c) VPN

Group 'C'

Attempt any TWO questions: **(2 × 10 = 20)**

9. Critically analyze the OSI reference mode. 10
10. Explain the random-access protocols under the multiple access taxonomy. 10
11. Explain the IPv4 Header format in detail. 10

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